

17. Given $C_1 = 2+j$, $C_2 = -2+j3$, calculate the following

(a) $C_1 + C_2$

(b) $C_1 - C_2$

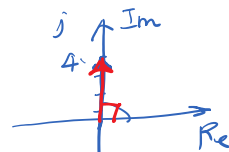
(c) $C_1 C_2$

(d) C_1 / C_2

give the answers in both rectangular and polar forms

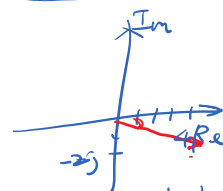
$$\begin{aligned} \text{a) } C_1 + C_2 &= 2+j + (-2+j3) \\ &= (2-2) + j(1+3) \\ &= j4 \end{aligned}$$

$$\begin{aligned} |C_1 + C_2| &= \sqrt{0^2 + 4^2} = 4 \\ \angle \theta &= \arctan \frac{4}{0} = \frac{\pi}{2} \end{aligned}$$



$$j4 = 4e^{j\frac{\pi}{2}}$$

$$\begin{aligned} \text{b) } C_1 - C_2 &= (2+j) - (-2+j3) \\ &= (2+2) + j(1-3) \\ &= 4 - j2 \end{aligned}$$

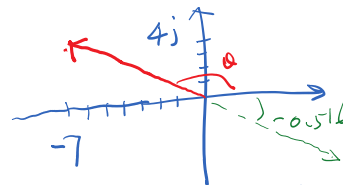


$$|C_1 - C_2| = \sqrt{4^2 + 2^2} = 4.472$$

$$\theta = \arctan \frac{-2}{4} = -0.462 \text{ radians}$$

$$\therefore C_1 - C_2 = 4.472 e^{-j0.462}$$

$$\begin{aligned} \text{c) } C_1 C_2 &= (2+j) \cdot (-2+j3) \\ &= 2 \cdot (-2) + 2 \cdot (j3) + j(-2) + j(j3) \\ &= -4 + j6 - j2 + 3j^2 \\ &= -4 + j4 + 3(-1) \\ &= -7 + j4 \end{aligned}$$



$$|C_1 C_2| = \sqrt{7^2 + 4^2} = 8.062$$

$$\theta = \arctan \frac{4}{-7} = -0.516$$

$$= -0.516 + \pi$$

$$= 2.623$$

$$\therefore C_1 C_2 = 8.062 e^{j2.623}$$

$$\text{d) } C_1 / C_2 = \frac{2+j}{-2+j3}$$

$$= \frac{2+j}{(-2+j3)(-2-j3)}$$

$$j^2 = -1$$

$$= 1.122 e^{j(\theta_1 + \theta_2)}$$

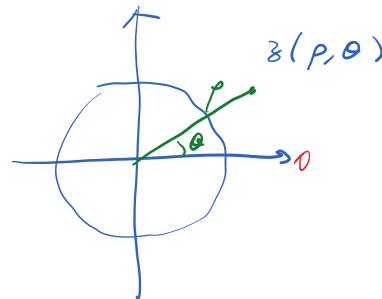
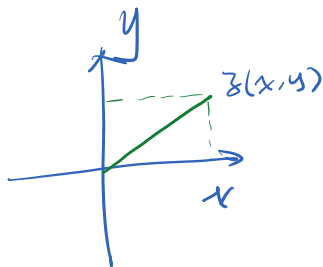
$$\begin{aligned}
 &= \frac{-}{(-2+j3)(-2-j3)} \\
 &= \frac{-(2+j)(2+j3)}{(-2)^2 - (j3)^2} \\
 &= \frac{-(4+6j+j^2+j^2 3)}{4 - j^2 \cdot 9} \\
 &= \frac{-(1+j8)}{4+9} \\
 &= -\frac{1}{13} - j\frac{8}{13} \\
 &= -0.0769 - 0.6154j
 \end{aligned}$$

$$\therefore C_1 C_2 = 8.062 e^{j2.6}$$

$$\begin{aligned}
 C_1 \cdot C_2 &= |C_1| |C_2| e^{j(\theta_1 + \theta_2)} \\
 C_1 / C_2 &= \frac{|C_1|}{|C_2|} e^{j(\theta_1 - \theta_2)}
 \end{aligned}$$

$$\begin{aligned}
 |z| &= \sqrt{(-0.0769)^2 + (-0.6154)^2} \\
 &= 0.62 \\
 \theta &= \arctan\left(\frac{-0.615}{-0.0769}\right) = \arctan(8) \\
 &= \arctan(7.999) = 1.4465 \\
 &= 1.4465 - \pi = -1.695 \\
 \therefore z &= 0.62 e^{j1.495}
 \end{aligned}$$

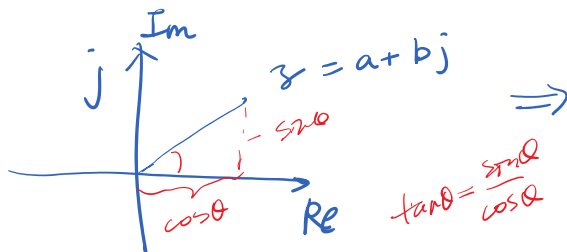
Now, How to transfer it from rectangular to polar form



$$\begin{aligned}
 p &= |z| = \sqrt{x^2 + y^2} \\
 \theta &= \arctan \frac{y}{x}
 \end{aligned}$$

$$\begin{aligned}
 &0 \sim 2\pi \\
 &0 \sim 360 \\
 &\text{degree} \cdot \frac{\pi}{180} = \text{radians}
 \end{aligned}$$

Similarly for complex number



$$\begin{aligned}
 z &= |z| \cdot \angle \theta \\
 &= |z| e^{j\theta}
 \end{aligned}$$

$$\begin{aligned}
 |z| &= \sqrt{a^2 + b^2} \\
 \angle \theta &= \arctan \left(\frac{b}{a} \right)
 \end{aligned}$$

$$\begin{aligned}
 z &= |z| e^{j\theta} \\
 &= |z| (\cos \theta + j \sin \theta) \\
 &= |z| \cos \theta + j |z| \sin \theta \\
 &= a + j b
 \end{aligned}$$

Note: the radius and ~~angles~~ degree

$$\omega \frac{\pi}{180^\circ} \theta \text{ degree}$$

$$\therefore \frac{\omega}{\pi} = \frac{\theta}{180^\circ}$$

$$\therefore \theta = \omega \cdot \frac{180^\circ}{\pi}$$

for example. radius 0.5 $\Rightarrow \frac{0.5 \cdot 180^\circ}{\pi} = 28.6^\circ$

2) See the four parts below, Give answers in rectangular form

a) Given the equation $z^2 = 1 + j$

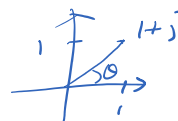
Find two values of z that satisfy this equation

$$z^2 = 1 + j$$

transfer from rectangle format to polar format

$$|1+j| = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\theta = \arctan\left(\frac{1}{1}\right) = \frac{\pi}{4}$$



$$\therefore 1+j = \sqrt{2} \cdot e^{j(\frac{\pi}{4} + 2\pi k)} \quad k = 0, \pm 1, \pm 2, \pm 3, \dots, \text{repeat?}$$

uniq value. $k = 0, 1$

when $k=0$: $1+j = \sqrt{2} e^{j\frac{\pi}{4}}$

$$\therefore z_1 = \sqrt[4]{\sqrt{2}} \cdot e^{j\frac{\pi}{8}}$$

$$= 1.189 e^{j\frac{\pi}{8}}$$

when $k=1$:

Real

$$x^2 = b$$

$$x = \pm \sqrt{b}$$

$$1+j = \sqrt{2} \cdot e^{j(\frac{\pi}{4} + 2\pi)}$$

$$\begin{aligned} \therefore z_2 &= \sqrt{2} \cdot e^{j(\frac{\pi}{4} + 2\pi)/2} \\ &= 1.189 \cdot e^{j(\frac{\pi}{8} + \pi)} \\ &= 1.189 e^{j\frac{9\pi}{8}} \end{aligned}$$



Note: $z_1 = -z_2$

write on rectangle format

$$\begin{aligned} z_1 &= 1.189 e^{j\frac{1}{8}\pi} = 1.189 (\cos \frac{1}{8}\pi + j \sin \frac{1}{8}\pi) \\ &= 1.189 (0.9238 + j 0.3826) \\ &= 1.098 + j 0.455 \end{aligned}$$

Similarly

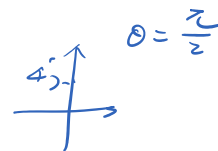
$$\begin{aligned} z_2 &= 1.189 e^{j\frac{9}{8}\pi} = 1.189 (\cos \frac{9}{8}\pi + j \sin \frac{9}{8}\pi) \\ &= -1.098 - j 0.455 \end{aligned}$$

There are more select of k , but they will have the same value of either z_1 or z_2

b) Calculate $(4j)^{\frac{1}{2}}$

First, transfer to polar format

$$\begin{aligned} 4j &\rightarrow |4j| \arctan \frac{4}{0} \\ &= 4 e^{j\frac{\pi}{2}} \end{aligned}$$



$$\therefore (4j)^{\frac{1}{2}} = \sqrt{4} \cdot e^{j(\frac{\pi}{2} + 2k\pi)/2}, \quad k = 0, \pm 1, \pm 2, \dots$$

$$k=0: z_1 = (4j)^{\frac{1}{2}} = 2 \cdot e^{j\frac{\pi}{4}}$$

$$k=1: z_2 = (4j)^{\frac{1}{2}} = 2 \cdot e^{j\frac{\pi}{4} + \pi}$$

write transfer to polar format

$$e^{j\theta} = \cos\theta + j \sin\theta$$

$$\begin{aligned} z_1 &= 2 \cdot e^{j\frac{\pi}{4}} = 2 \cdot \left(\cos\left(\frac{\pi}{4}\right) + j \sin\left(\frac{\pi}{4}\right) \right) \\ &= 2 \cdot \left(\frac{\sqrt{2}}{2} + j \frac{\sqrt{2}}{2} \right) \\ &= \sqrt{2} + j\sqrt{2} \end{aligned}$$

When $k=1$.

$$\begin{aligned} z_2 &= (4j)^{\frac{1}{2}} = \sqrt{4} \cdot e^{j(\frac{\pi}{4} + \pi)} \\ &= 2 \cdot e^{j\frac{5\pi}{4}} \end{aligned}$$

$$\begin{aligned} &= 2 \cdot \left[\cos\left(\frac{5\pi}{4}\right) + j \sin\left(\frac{5\pi}{4}\right) \right] \\ &= 2 \cdot \left[-\frac{\sqrt{2}}{2} - j \frac{\sqrt{2}}{2} \right] \\ &= -\sqrt{2} - j\sqrt{2} \end{aligned}$$



Note $z_1 = -z_2$

c) Calculate $\sqrt{4j}$

Note: $4j$ transfer to polar format

$$4j = 4e^{j\frac{\pi}{2}}$$

$$\begin{aligned}\therefore \sqrt{4j} &= \sqrt{4e^{j\frac{\pi}{2}}} \\ &= \sqrt{4} \cdot \sqrt{e^{j\frac{\pi}{2}}} \\ &= 2 \cdot e^{j\frac{\pi}{4}} \\ &= \sqrt{2} + j\sqrt{2}\end{aligned}$$

d) calculate $(1+2j)^3$

Step. transfer $1+2j$ to polar format.

$$|1+2j| = \sqrt{1^2 + 2^2} = \sqrt{5}$$

$$\theta = \arctan\left(\frac{2}{1}\right) = \dots$$

$$\begin{aligned}\text{the } (1+2j)^3 &= (\sqrt{5})^3 + \theta^3 \\ &= \dots \\ &= \dots\end{aligned}$$

No, we can calculate it directly

$$\begin{aligned}(1+2j)^3 &= (1+2j)^2 \cdot (1+2j) \\ &= (1+4j+4j^2)(1+2j) \\ &= (-3+4j)(1+2j)\end{aligned}$$

$$= -3 - 6j + 4j + 8j^2$$

$$= -11 - j^2$$

3) Give the complex number that corresponding to the phasor representation of the following harmonic functions

a) $F(t) = 3 \sin(\omega t - \pi/2)$

time domain to phasor domain
write down the format of

Constant Value. $\cos(\omega t - \theta)$

Same $\omega \Rightarrow$ harmonic.

Note $e^{j\omega t - \theta}$

$$= \cos(\omega t - \theta) + j \sin(\omega t - \theta)$$

So, I can just $\text{Re}[e^{j\omega t - \theta}]$

$$F(t) = 3 \sin(\omega t - \pi/2)$$

Note $\cos \theta = \sin(\frac{\pi}{2} - \theta)$

$$= 3 \cdot \cos\left(\frac{\pi}{2} - (\omega t - \frac{\pi}{2})\right)$$

$$= 3 \cdot \cos(\pi - \omega t)$$

note.

$$\cos \theta = \cos(-\theta)$$

$$= \underline{3} \cos(\omega t - \pi)$$

$$\therefore F(t) = \text{Re} \left[3 \cdot e^{j(\omega t - \pi)} \right]$$

$$= \text{Re} \left[3 \cdot (\cos \omega t - \pi) + j \sin(\omega t - \pi) \right]$$

imaginary part eliminated.

∴ Amplitude is $A=3$.

phase angle $\phi = -\pi$

phasor F is represented as $Ae^{j\phi}$

$$\text{Note } e^{j(\omega t + \phi)} = \underline{e^{j\omega t}} \cdot e^{j\phi}$$

$$\begin{aligned} \therefore F(t) &= \operatorname{Re} \{ 3 e^{j\omega t} e^{-j\pi} \} \\ &= \operatorname{Re} \{ \underbrace{3 \cdot e^{-j\pi}}_{\text{phasor}} e^{j\omega t} \} \end{aligned}$$

∴ The phasor of $F(t)$ is $\underline{F} = 3 e^{-j\pi}$

Use Euler's formula $e^{j\theta} = \cos\theta + j\sin\theta$.

$$\begin{aligned} \therefore \underline{F} &= 3(\cos(-\pi) + j\sin(-\pi)) \\ &= 3(-1 + j0) \\ &= -3 \end{aligned}$$



$$b) \quad G(t) = -2 \cos(\omega t)$$

So $\cos(\omega t) = e^{j\omega t}$, take real part only

$$= \cos\omega t + j\sin\omega t \quad \text{take real part only}$$

$$\therefore \cos(\omega t) = \operatorname{Re} \{ e^{j\omega t} \}$$

$$\therefore G(t) = -2 \cos(\omega t) \\ = \text{Re} \left\{ \underbrace{-2 \cdot e^{j\omega t}}_{\text{phasor}} \right\}$$

$\therefore$ phasor is -2

$$\begin{aligned} \text{c) } H(t) &= 4 \cos\left(60\pi t - \frac{\pi}{4}\right) \\ &= 4 \left[\underbrace{\cos\left(60\pi t - \frac{\pi}{4}\right) + j \sin\left(60\pi t - \frac{\pi}{4}\right)}_{e^{j\left(60\pi t - \frac{\pi}{4}\right)}} \right] \cdot \begin{matrix} \nwarrow \\ \text{take real} \\ \text{part only} \end{matrix} \\ &= \text{Re} \left\{ 4 e^{j\left(60\pi t - \frac{\pi}{4}\right)} \right\} \\ &= \text{Re} \left\{ 4 e^{j60\pi t} e^{-j\frac{\pi}{4}} \right\} \quad \omega = 60\pi \\ &= \text{Re} \left\{ \underbrace{4 \cdot e^{-j\frac{\pi}{4}}}_{\text{phasor}} \cdot e^{j60\pi t} \right\} \end{aligned}$$

$$\begin{aligned} \text{So, the phasor is } &4 e^{-j\frac{\pi}{4}} \\ &= 4 \left(\cos\left(\frac{\pi}{4}\right) + j \sin\left(\frac{\pi}{4}\right) \right) \\ &= 4 \left(\frac{\sqrt{2}}{2} - j \frac{\sqrt{2}}{2} \right) \\ &= 2\sqrt{2} - j2\sqrt{2} \end{aligned}$$

$$d) E(t) = 4 \sin \omega t + \cos \omega t.$$

Now we have both $\sin$ & $\cos$. we need all in $\cos$ form. $\sin \theta = \cos(\frac{\pi}{2} - \theta)$

$$E(t) = 4 \cos(\frac{\pi}{2} - \omega t) + \cos(\omega t)$$

$$\text{Let } E_1(t) = 4 \cos(\frac{\pi}{2} - \omega t) = 4 \cos(\omega t - \frac{\pi}{2})$$

$$E_2(t) = \cos(\omega t)$$

Same ω , so the phasor can be added.

$$E_1(t) = \text{Re} \left\{ 4 \cdot \left[\cos(\frac{\pi}{2} - \omega t) - j \sin(\frac{\pi}{2} - \omega t) \right] \right\}$$

$$= \text{Re} \left\{ 4 \cdot e^{j(\omega t - \frac{\pi}{2})} \right\}$$

$$= \text{Re} \left\{ \underbrace{4 \cdot e^{-j\frac{\pi}{2}}}_{\text{phasor}} e^{j\omega t} \right\}$$

$$E_2(t) = \cos \omega t$$

$$= \text{Re} \left\{ \cos(\omega t) + j \sin(\omega t) \right\}$$

$$= \text{Re} \left\{ \underbrace{1}_{\uparrow} \cdot e^{j\omega t} \right\}$$

$$\text{phasor} = 1$$

$$\therefore \text{phase of } \underbrace{E(t)}_{E_1} = \underbrace{4e^{-j\frac{\pi}{2}}}_{E_2} + \underbrace{1}_{E_3}$$



$$= -4j + 1$$

4) Obtain the sinusoidal y-varying function $K(t)$ corresponding to the following phasor.

a) $K = 1 - j$

time $K(t)$ ① Write to wave take Re part
② with $\cos + j \sin$ separate $e^{j\omega t}$, get phasor phasor K .

$$F(t) = \text{Re} \left\{ \underbrace{A e^{j\phi}}_{\text{phasor}} e^{j\omega t} \right\}$$

$$K = 1 - j \Rightarrow \text{Polar } K = \sqrt{2} e^{-j\frac{\pi}{4}}$$



$$\begin{aligned} K(t) &= \text{Re} \left\{ K e^{j\omega t} \right\} \\ &= \text{Re} \left\{ (1-j) e^{j\omega t} \right\} \\ &= \text{Re} \left\{ \sqrt{2} e^{-j\frac{\pi}{4}} e^{j\omega t} \right\} \\ &= \text{Re} \left\{ \sqrt{2} \cdot e^{j(\omega t - \frac{\pi}{4})} \right\} \end{aligned}$$

$$= \operatorname{Re} \left\{ \sqrt{2} \cos(\omega t - \frac{\pi}{4}) + j \sin(\omega t - \frac{\pi}{4}) \right\}$$

$$= \sqrt{2} \cos(\omega t - \frac{\pi}{4})$$

b) $K = e^{j0.4}$

this is already in polar form, good,

$$k(t) = \operatorname{Re} \{ K e^{j\omega t} \}$$

$$= \operatorname{Re} \{ e^{j0.4} \cdot e^{j\omega t} \}$$

$$= \operatorname{Re} \{ e^{j(\omega t + 0.4)} \}$$

$$= \operatorname{Re} \{ \cos(\omega t + 0.4) + j \sin(\omega t + 0.4) \}$$

$$= \cos(\omega t + 0.4)$$

c) $K = 2e^{-j\frac{\pi}{4}} + 3e^{j0.3}$

$$k(t) = \operatorname{Re} \{ K e^{j\omega t} \}$$

Note, K is complex, so we are multiply two complex numbers.

Use polar form is easier.

K is two complex addition, so use rectangular form is easier.

$$\begin{aligned}
 K &= 2e^{-j\frac{\pi}{4}} + 3e^{j0.3} \\
 &= 2\left[\cos\left(-\frac{\pi}{4}\right) + j\sin\left(-\frac{\pi}{4}\right)\right] + 3(\cos(0.3) + j\sin(0.3)) \\
 &= 2\left(\frac{\sqrt{2}}{2} + j\left(-\frac{\sqrt{2}}{2}\right)\right) + (2.866 + 0.877j) \\
 &= (1.414 + 2.866) + (-1.414 + 0.877j) \\
 &= 4.280 - 0.528j
 \end{aligned}$$

Okay, Now do the multiplication, use polar format:

$$|K| = \sqrt{4.28^2 + (-0.528)^2} = 4.313$$

$$\angle K = \arctan\left(\frac{-0.528}{4.28}\right) = -7.03^\circ$$

$$\therefore \text{In polar format, } K = 4.313 \angle -7.03^\circ$$

$$\therefore K(t) = \operatorname{Re} \{ K \cdot e^{j\omega t} \}$$

$$= \operatorname{Re} \{ 4.313 \cdot e^{-j7.03^\circ} \cdot e^{j\omega t} \}$$

$$= \operatorname{Re} \{ 4.313 e^{j(\omega t - 7.03^\circ)} \}$$

$$= \operatorname{Re} \{ 4.313 \cos(\omega t - 7.03^\circ) + j \sin(\omega t - 7.03^\circ) \}$$

$$= 4.313 \cos(\omega t - 7.03^\circ)$$

degree radians
 -7.03° -0.123

$$= -4.313 \cos(\omega t - 0.123)$$

$$7.05 \leftarrow \dots$$

$$7.05^\circ \times \frac{\pi}{180} = 0.123$$

12.5 Let $\underline{A} = \underline{\hat{x}} - 2\underline{\hat{y}} + 3\underline{\hat{z}}$

$$\underline{B} = -\underline{\hat{x}} - 2\underline{\hat{y}} + 3\underline{\hat{z}}$$

a) Find $\underline{A} + \underline{B}$

b) Find $\underline{A} - \underline{B}$

c) Find $\underline{A} \cdot \underline{B}$

d) Find $\underline{A} \times \underline{B}$

Solution

a)

$$\begin{aligned} \underline{A} + \underline{B} &= (\underline{\hat{x}} - 2\underline{\hat{y}} + 3\underline{\hat{z}}) + (-\underline{\hat{x}} - 2\underline{\hat{y}} + 3\underline{\hat{z}}) \\ &= (0\underline{\hat{x}} - 4\underline{\hat{y}} + 6\underline{\hat{z}}) \\ &= -4\underline{\hat{y}} + 6\underline{\hat{z}} \end{aligned}$$

b)

$$\begin{aligned} \underline{A} - \underline{B} &= (\underline{\hat{x}} - (-\underline{\hat{x}})) + (-2 + 2)\underline{\hat{y}} + (3 - 3)\underline{\hat{z}} \\ &= 2\underline{\hat{x}} \end{aligned}$$

c)

$$\begin{aligned} \underline{A} \cdot \underline{B} &= \underline{\hat{x}} \cdot (-2\underline{\hat{x}}) + (-2\underline{\hat{y}}) \cdot (-2\underline{\hat{y}}) + (3\underline{\hat{z}}) \cdot (3\underline{\hat{z}}) \\ &= -1 \cdot \underbrace{|\underline{\hat{x}}|^2}_1 + 4 + 9 \end{aligned}$$

$$= 12$$

$$c) \underline{A} \times \underline{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 1 & -2 & 3 \\ -1 & -2 & 3 \end{vmatrix}$$

$$= \hat{x} \begin{vmatrix} -2 & 3 \\ -2 & 3 \end{vmatrix} - \hat{y} \begin{vmatrix} 1 & 3 \\ -1 & 3 \end{vmatrix} + \hat{z} \begin{vmatrix} 1 & -2 \\ -1 & -2 \end{vmatrix}$$

$$= \hat{x} \cdot (-6 - (-6)) - \hat{y} (3 - (-3)) + \hat{z} (-2 - (2))$$

$$= 0 - 6\hat{y} - 4\hat{z}$$

$$= -6\hat{y} - 4\hat{z}$$

*6 Find the complex phasor vector $\underline{A}$ for the following time-harmonic vector.

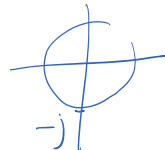
$$(a) \underline{A}(t) = \cos(\omega t) \hat{x} - 2\cos(\omega t - \frac{\pi}{2}) \hat{y} - 2\sin(\omega t) \hat{z}$$

Solution. find the phasor for each components. $\hat{x}$. $\hat{y}$. $\hat{z}$

$$A_x = \cos(\omega t) = 1 \cdot \cos(\omega t + 0) \Rightarrow 1 \cdot e^{j0}$$

$$\text{Find the phasor. } \tilde{A}_x = 1$$

Find the phasor. $\tilde{A}_x = 1$

$$A_y = -2 \cos(\omega t - \frac{\pi}{2}) \Rightarrow -2 e^{j(-\frac{\pi}{2})} = -2 \cdot (-j) = 2j$$


$$\begin{aligned} A_z &= -2 \sin(\omega t) \\ &= -2 \cos(\frac{\pi}{2} - \omega t) \\ &= -2(\omega t - \frac{\pi}{2}) \Rightarrow -2 e^{j(-\frac{\pi}{2})} = -2(-j) = 2j \end{aligned}$$

Complex phasor of vector $\hat{A}$:

$$\underline{\hat{A}} = 1 \cdot \underline{\hat{x}} + (2j) \underline{\hat{y}} + (2j) \underline{\hat{z}}$$

Rule to remember

$$\begin{aligned} \cos(\omega t + \phi) &\Rightarrow e^{j\phi} \\ \sin(\omega t + \phi) &\Rightarrow -j e^{j\phi} \end{aligned}$$

$$A(t) = (\sin \omega t - 3 \sin \omega t) \underline{\hat{y}} + (\sin \omega t - \cos \omega t) \underline{\hat{z}}$$

$$\begin{aligned} A_y &= (\sin \omega t - 3 \sin \omega t) \\ &= -2 \sin \omega t \end{aligned} \quad \begin{aligned} \hat{A}_y \text{ phasor} &= \\ &\Rightarrow -2 \cdot (-j e^{j0}) \\ &= 2j \cdot 1 = 2j \end{aligned}$$

$$\begin{aligned} A_z &= (\sin \omega t - \cos \omega t) \\ &\Rightarrow -j e^{j0} - e^{j0} \\ &= -j - 1 \end{aligned}$$

$\therefore$ The complex phasor of vector is

$$\underline{\hat{A}} = 2j \underline{\hat{y}} + (-j - 1) \underline{\hat{z}}$$

$$c) A(t) = 0.3 \sin(ky - \omega t) \hat{x}$$

Note, here, there are only A_x component in $\hat{x}$ direction

$$A_x = 0.3 \sin(ky - \omega t)$$

it is a function of y , not the $\hat{y}$.

$$A_x = 0.3 \sin(ky - \omega t)$$

$$= -0.3 \sin(\omega t - ky)$$

Note:

$$\sin(\omega t + \phi) \Rightarrow -je^{j\phi}$$

(phasor)

$$\tilde{A}_x = -0.3(-j)e^{j(-ky)}$$

$$= 0.3j \cdot e^{-jky}$$

$$= 0.3 e^{j\frac{\pi}{2}} \cdot e^{-jky}$$

$$= 0.3 e^{j(\frac{\pi}{2} - ky)}$$

Note $j = e^{j\frac{\pi}{2}}$

~~The~~ complex phasor of $\underline{A}(t)$ is

$$\underline{\tilde{A}} = 0.3 e^{j(\frac{\pi}{2} - ky)} \hat{x}$$

$$\text{or } = 0.3 j e^{jky} \hat{x} \text{ is fine.}$$

#7. From the following complex phasor vector,
find the sinusoidal vector $k(t)$ in terms of ωt .

This is asking for transfer from phasor domain to time domain

$$\underset{\text{phasor}}{\tilde{F}} \xrightarrow{\text{time}} \underset{\text{phasor}}{\text{Re} \{ \tilde{F} \cdot e^{j\omega t} \}}$$

$$a) \underline{k} = j \hat{y} + \hat{z}$$

Again, transfer one by one.

$$\begin{aligned} j &\xrightarrow{\text{time}} \text{Re} \{ j \cdot e^{j\omega t} \} \\ &= \text{Re} \{ j (\cos \omega t + j \sin \omega t) \} \\ &= \text{Re} \{ j \cos \omega t - \sin \omega t \} \\ &= -\sin \omega t \end{aligned}$$

$$\begin{aligned} 1 &\xrightarrow{\text{time}} \text{Re} \{ 1 \cdot e^{j\omega t} \} \\ &= \text{Re} \{ 1 \cdot (\cos \omega t + j \sin \omega t) \} \\ &= \cos \omega t \end{aligned}$$

Thus, the time domain

$$K(t) = -\sin(\omega t) \hat{x} + \cos(\omega t) \hat{y}$$

some time we prefer to in cos format then

$$\begin{aligned} K(t) &= -\cos\left(\frac{\pi}{2} - \omega t\right) \hat{x} + \cos(\omega t) \hat{y} \\ &= \cos\left(\frac{\pi}{2} - \omega t + \pi\right) \hat{x} + \cos(\omega t) \hat{y} \\ &= \cos\left(\omega t - \frac{3\pi}{2} + 2\pi\right) \hat{x} + \cos(\omega t) \hat{y} \\ &= \cos\left(\omega t + \frac{\pi}{2}\right) \hat{x} + \cos(\omega t) \hat{y} \end{aligned}$$

$$b) \underline{K} = j(\hat{y} + j\hat{x})$$

First, simplify it

$$\begin{aligned} K &= j\hat{y} + j^2\hat{x} \\ &= j\hat{y} - \hat{x} \end{aligned}$$

From a), we can see that

$$K(t) = \cos\left(\omega t + \frac{\pi}{2}\right) \hat{x} - \cos(\omega t) \hat{y}$$

$$c) \underline{K} = e^{-jky} \hat{x} - j e^{-jky} \hat{y}$$

$$K_x = e^{-jky}$$

transfer to time domain

$$\begin{aligned} K_x(t) &= \text{Re} \{ e^{-jky} e^{j\omega t} \} \\ &= \text{Re} \{ e^{j(\omega t - ky)} \} \\ &= \cos(\omega t - ky) \end{aligned}$$

$$\begin{aligned} K_z(t) &= \text{Re} \{ -j e^{-jky} \cdot e^{j\omega t} \} \\ &= \text{Re} \{ e^{-j\frac{\pi}{2}} \cdot e^{-jky} \cdot e^{j\omega t} \} \\ &= \text{Re} \{ e^{j(\omega t - ky - \frac{\pi}{2})} \} \\ &= \cos(\omega t - ky - \frac{\pi}{2}) \end{aligned}$$

Note
 $-j = e^{-j\frac{\pi}{2}}$

$$\therefore K(t) = \cos(\omega t - ky) \underline{\hat{x}} + \cos(\omega t - ky - \frac{\pi}{2}) \underline{\hat{z}}$$

Alternative solution

$$K(t) = \cos(\omega t - ky) \underline{\hat{x}} + \sin(\omega t - ky) \underline{\hat{z}}$$

Note. $\text{Re}\{e^{j\theta}\} = \cos \theta$
 $\text{Re}\{-j e^{j\theta}\} = \sin \theta$

* problem 8

Sketch the trace of the tip of $A(t)$ in the xy plane.
 where the phasor are:

a) $A = 3\bar{x} + 3\bar{y}$

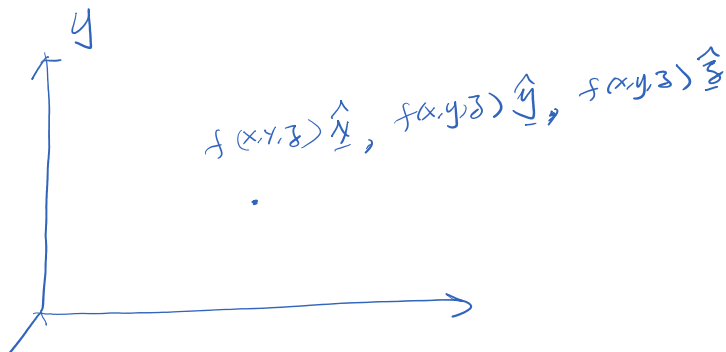
So the phasor is given to us, need to sketch in time domain.

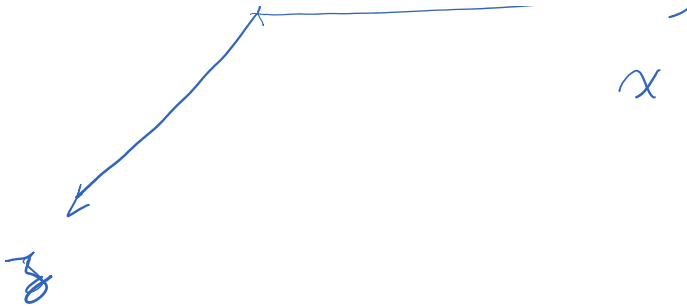
so the first step is to transfer phasor to time domain

$$3 \xrightarrow{\text{time}} \text{Re}\{3 \cdot e^{j\omega t}\} = 3 \cos \omega t$$

$$3 \longrightarrow 3 \cos \omega t$$

$$\therefore A(t) = \cos(\omega t) \hat{x} + \cos \omega t \hat{y}$$





$$f(x, y, z) = \cos(\omega t)$$

$$f(x, y, z) = \cos(\omega t)$$

ECE 3317 – Applied Electromagnetic Waves Homework #1 – Solutions

Problem 1. Given $c_1 = 2 + j$ and $c_2 = -2 + 3j$.

In polar form: $c_1 = \sqrt{5} e^{j 26.57^\circ}$, $c_2 = \sqrt{13} e^{j 123.69^\circ}$

(a) $c_1 + c_2 = (2 + j) + (-2 + 3j) = 0 + 4j$

$$c_1 + c_2 = 4j = 4 e^{j 90^\circ}$$

(b) $c_1 - c_2 = (2 + j) - (-2 + 3j) = 4 - 2j$

$$c_1 - c_2 = 4 - 2j = \sqrt{20} e^{-j 26.57^\circ} = 2\sqrt{5} e^{-j 26.57^\circ}$$

(c) $c_1 c_2 = (2 + j)(-2 + 3j) = -4 + 6j - 2j + 3j^2 = -4 + 4j - 3 = -7 + 4j$

$$c_1 c_2 = -7 + 4j = \sqrt{65} e^{j 150.26^\circ}$$

(d) Multiply numerator and denominator by the conjugate of c_2 :

$$\frac{c_1}{c_2} = \frac{(2 + j)(-2 - 3j)}{(-2 + 3j)(-2 - 3j)} = \frac{-4 - 6j - 2j - 3j^2}{4 + 9} = \frac{-1 - 8j}{13}$$

$$\frac{c_1}{c_2} = -\frac{1}{13} - \frac{8}{13}j = \frac{\sqrt{5}}{\sqrt{13}} e^{-j 97.13^\circ}$$



HW1

Audio recording
started: 3:40
PM Friday,
February 27,
2026

Problem 2. Give answers in rectangular form.

(a) $z^2 = 1 + j$. Convert to polar: $1 + j = \sqrt{2} e^{j\pi/4}$.

$$z = (\sqrt{2})^{1/2} e^{j\pi/8 + jn\pi}, \quad n = 0, 1$$

$$z_1 = 2^{1/4} e^{j\pi/8} = 2^{1/4} (\cos 22.5^\circ + j \sin 22.5^\circ)$$

$$\boxed{z_1 \approx 1.0987 + 0.4551j, \quad z_2 \approx -1.0987 - 0.4551j}$$

(b) $(4j)^{1/2}$. Write $4j = 4 e^{j\pi/2}$.

$$(4j)^{1/2} = 2 e^{j\pi/4} = 2 \left(\frac{\sqrt{2}}{2} + j \frac{\sqrt{2}}{2} \right)$$

$$\boxed{(4j)^{1/2} = \sqrt{2} + j\sqrt{2} \approx 1.414 + 1.414j}$$

(The other root is $-\sqrt{2} - j\sqrt{2}$.)

(c) $(1 + 2j)^3$. First compute $(1 + 2j)^2$:

$$(1 + 2j)^2 = 1 + 4j + 4j^2 = -3 + 4j$$

$$(1 + 2j)^3 = (-3 + 4j)(1 + 2j) = -3 - 6j + 4j + 8j^2 = -11 - 2j$$

$$\boxed{(1 + 2j)^3 = -11 - 2j}$$

Problem 3. Find the phasor for the following time-harmonic functions.

Convention: $f(t) = \text{Re}[\mathbf{F} e^{j\omega t}]$. Key rules: $\cos(\omega t + \phi) \rightarrow e^{j\phi}$, $\sin(\omega t + \phi) \rightarrow -j e^{j\phi}$.

(a) $F(t) = 3 \sin(\omega t - \pi/2)$

Using $\sin(\omega t - \pi/2) = -\cos(\omega t)$:

$$\boxed{\mathbf{F} = -3}$$

(b) $G(t) = -2 \cos(\omega t)$

$$\boxed{\mathbf{G} = -2}$$

(c) $H(t) = 4 \cos(60\pi t - \pi/4)$

Here $\omega = 60\pi$. The phasor is:

$$\mathbf{H} = 4 e^{-j\pi/4} = 4 \left(\frac{\sqrt{2}}{2} - j \frac{\sqrt{2}}{2} \right)$$

$$\boxed{\mathbf{H} = 4 e^{-j\pi/4} = 2\sqrt{2} - j2\sqrt{2}}$$

(d) $E(t) = 4 \sin \omega t + \cos \omega t$

$$4 \sin \omega t \rightarrow -4j, \quad \cos \omega t \rightarrow 1$$

$$\boxed{\mathbf{E} = 1 - 4j}$$

Problem 4. Find $K(t) = \text{Re}[\mathbf{K} e^{j\omega t}]$ for the given phasors.

(a) $\mathbf{K} = 1 - j = \sqrt{2} e^{-j\pi/4}$

$$K(t) = \text{Re}[(1 - j)(\cos \omega t + j \sin \omega t)] = \cos \omega t + \sin \omega t$$

$$\boxed{K(t) = \sqrt{2} \cos(\omega t - \pi/4)}$$

(b) $\mathbf{K} = e^{j0.4}$

$$\boxed{K(t) = \cos(\omega t + 0.4)}$$

(c) $\mathbf{K} = 2e^{-j\pi/4} + 3e^{j0.3}$

Convert each term to rectangular form:

$$2e^{-j\pi/4} = 2 \cos(-\pi/4) + j 2 \sin(-\pi/4) = \sqrt{2} - j\sqrt{2} \approx 1.414 - 1.414 j$$

$$3e^{j0.3} = 3 \cos(0.3) + j 3 \sin(0.3) \approx 2.866 + 0.887 j$$

$$\mathbf{K} = (1.414 + 2.866) + j(-1.414 + 0.887) = 4.280 - 0.528 j$$

$$|\mathbf{K}| = \sqrt{4.280^2 + 0.528^2} = 4.313, \quad \angle \mathbf{K} = \arctan\left(\frac{-0.528}{4.280}\right) = -7.03^\circ$$

$$\boxed{K(t) = 4.280 \cos \omega t + 0.528 \sin \omega t = 4.313 \cos(\omega t - 7.03^\circ)}$$

Problem 5. Let $\underline{A} = \hat{x} - 2\hat{y} + 3\hat{z}$ and $\underline{B} = -\hat{x} - 2\hat{y} + 3\hat{z}$.

(a)

$$\underline{A} + \underline{B} = (1 - 1)\hat{x} + (-2 - 2)\hat{y} + (3 + 3)\hat{z} = -4\hat{y} + 6\hat{z}$$

(b)

$$\underline{A} - \underline{B} = (1 + 1)\hat{x} + (-2 + 2)\hat{y} + (3 - 3)\hat{z} = 2\hat{x}$$

(c)

$$\underline{A} \cdot \underline{B} = (1)(-1) + (-2)(-2) + (3)(3) = -1 + 4 + 9 = 12$$

(d)

$$\underline{A} \times \underline{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 1 & -2 & 3 \\ -1 & -2 & 3 \end{vmatrix}$$

$$\begin{aligned} &= \hat{x}[(-2)(3) - (3)(-2)] - \hat{y}[(1)(3) - (3)(-1)] + \hat{z}[(1)(-2) - (-2)(-1)] \\ &= \hat{x}[-6 + 6] - \hat{y}[3 + 3] + \hat{z}[-2 - 2] \end{aligned}$$

$$\underline{A} \times \underline{B} = -6\hat{y} - 4\hat{z}$$

Problem 6. Find the complex phasor vector $\underline{\mathbf{A}}$ for the following time-harmonic vectors.

Rules: $\cos(\omega t + \phi) \rightarrow e^{j\phi}$, $\sin(\omega t + \phi) \rightarrow -j e^{j\phi}$.

(a) $\underline{\mathbf{A}}(t) = \cos(\omega t) \hat{x} - 2 \cos(\omega t - \pi/2) \hat{y} - 2 \sin(\omega t) \hat{z}$

$$\begin{aligned}\cos(\omega t) &\rightarrow 1 \\ -2 \cos(\omega t - \pi/2) &\rightarrow -2 e^{-j\pi/2} = -2(-j) = 2j \\ -2 \sin(\omega t) &\rightarrow -2(-j) = 2j\end{aligned}$$

$$\underline{\mathbf{A}} = \hat{x} + 2j \hat{y} + 2j \hat{z}$$

(b) $\underline{\mathbf{A}}(t) = (\sin \omega t - 3 \sin \omega t) \hat{y} + (\sin \omega t - \cos \omega t) \hat{z} = -2 \sin \omega t \hat{y} + (\sin \omega t - \cos \omega t) \hat{z}$

$$\begin{aligned}-2 \sin \omega t &\rightarrow -2(-j) = 2j \\ \sin \omega t - \cos \omega t &\rightarrow -j - 1\end{aligned}$$

$$\underline{\mathbf{A}} = 2j \hat{y} + (-1 - j) \hat{z}$$

(c) $\underline{\mathbf{A}}(t) = 0.3 \sin(ky - \omega t) \hat{x}$

Note: $\sin(ky - \omega t) = -\sin(\omega t - ky)$. Since $\sin(\omega t - ky) = \text{Re}[-j e^{-jky} e^{j\omega t}]$, we have:

$$-\sin(\omega t - ky) = \text{Re}[j e^{-jky} e^{j\omega t}]$$

$$\underline{\mathbf{A}} = 0.3j e^{-jky} \hat{x}$$

Problem 7. Find $\underline{K}(t) = \text{Re}[\underline{K} e^{j\omega t}]$.

(a) $\underline{K} = j \hat{y} + \hat{z}$

$$\begin{aligned}\underline{K}(t) &= \text{Re}[(j \hat{y} + \hat{z})(\cos \omega t + j \sin \omega t)] \\ &= \text{Re}[(j \cos \omega t - \sin \omega t) \hat{y} + (\cos \omega t + j \sin \omega t) \hat{z}]\end{aligned}$$

$$\boxed{\underline{K}(t) = -\sin \omega t \hat{y} + \cos \omega t \hat{z}}$$

(b) $\underline{K} = j(\hat{y} + j\hat{z}) = j \hat{y} - \hat{z}$

$$\underline{K}(t) = \text{Re}[(j \hat{y} - \hat{z})e^{j\omega t}]$$

$$\boxed{\underline{K}(t) = -\sin \omega t \hat{y} - \cos \omega t \hat{z}}$$

(c) $\underline{K} = e^{-jky} \hat{x} - j e^{-jky} \hat{z}$

$$\underline{K}(t) = \text{Re}[e^{j(\omega t - ky)} \hat{x} - j e^{j(\omega t - ky)} \hat{z}]$$

Using $\text{Re}[e^{j\theta}] = \cos \theta$ and $\text{Re}[-j e^{j\theta}] = \sin \theta$:

$$\boxed{\underline{K}(t) = \cos(\omega t - ky) \hat{x} + \sin(\omega t - ky) \hat{z}}$$

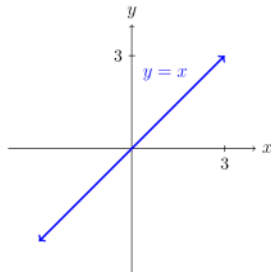
Problem 8. Sketch the trace of the tip of $\underline{A}(t)$ in the xy plane.

(a) $\underline{A} = 3\hat{x} + 3\hat{y}$ (both components real)

$$\underline{A}(t) = 3 \cos \omega t \hat{x} + 3 \cos \omega t \hat{y}$$

$$\text{So } x = 3 \cos \omega t, y = 3 \cos \omega t \Rightarrow y = x.$$

Linear polarization along the line $y = x$, from $(-3, -3)$ to $(3, 3)$.

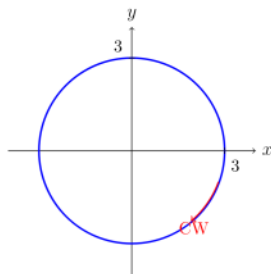


(b) $\underline{A} = 3\hat{x} + j3\hat{y}$

$$\underline{A}(t) = 3 \cos \omega t \hat{x} - 3 \sin \omega t \hat{y}$$

$$x = 3 \cos \omega t, y = -3 \sin \omega t \Rightarrow x^2 + y^2 = 9.$$

Circular polarization (radius 3, clockwise).



(c) $\underline{A} = 2\hat{x} + j3\hat{y}$

$$\underline{A}(t) = 2 \cos \omega t \hat{x} - 3 \sin \omega t \hat{y}$$

$$x = 2 \cos \omega t, y = -3 \sin \omega t \Rightarrow \left(\frac{x}{2}\right)^2 + \left(\frac{y}{3}\right)^2 = 1.$$

Elliptical polarization (semi-axes $a = 2$, $b = 3$, clockwise).

